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GenAI

Generative AI which is a subset of artificial Intelligence which is designed to generate new content.

How GenAI is different from Traditional ML?

Machine learning
Deep Learning
NLP

→ Major focus was on understanding pattern, classifying data and also used to make predictions.

GenAI uses the pattern it has learnt and generates new content which resembles human created work.

GenAI was trained on huge amount of data and based on its learning it generates text, codes, images, audio video etc

How it works?

① Training

Models have been trained on huge amount of data

② Prompting
↳ when we question model it generates content

Large Language models [LLM's]

LLM's are specific type of GenAI model which is trained on huge amount of data to understand, summarize and generate human like text.

How LLM's are created?

① Pretraining
The model has been trained on huge amount of data

② Fine tuning
Model has been trained on specific dataset so it can excel at specific task.

③ RLHF ⟶ Reinforcement Learning from Human Feedback.
Human feedback helps in [illegible] done

comprising the instruction
by the model.

Basic Concepts

① Tokens → Chunks of characters
→ apple

② Context window → short term memory
Defines how much information a model process at a time.

③ parameters
weights in neural network
captures complex information

LLM models

① OpenAI → ChatGPT
② Google → Gemini / Gemma
③ Meta AI → Llama
④ Anthropic → Claude
⑤ perplexity
⑥ DeepSeek
⑦ Mistral.
⑧ Groq.

Prompt Engineering

Process of designing the instructions given to the model.

instruction

AI

Question → How we question the model is done using prompt Engineering

It is just like asking better question to get better Results

Q1 → Machine learning

Better way → Explain machine learning in 5 bullet points

Q1 → Linear Regression

→ Give me code to implement Linear Regression.

Types of prompt Engineering

① Zero shot prompting

Basic form of prompting where we give AI a task without any examples.

! . Translate Good morning

→ [illegible]
into German Language

② Few shot prompting
→ we provide few example (shots) to the model.
→ used when we expect output in specific format.

Eg: Great Movie → positive sentiment

The movie was boring → Negative sentiment

The movie was just okay → Negative sentiment

③ Chain of thought (COT)
Instead of asking for direct answer, we instruct AI to think "step-by-step".

✗ Check whether 2 is prime or not

✓ Explain step by step how to find whether 2 is prime or not using Python

④ Role prompting (Persona)

By assigning specific role or persona to the AI so it influence its tone, expertise level and perspective.

Eg:- You are Senior Datascientist explain how to do projects in real time for beginner.

LLM API's

API → Application programming Interface.

API is set of rules the allows one software application to interact with other.